

THE UNIVERSITY OF NEW SOUTH WALES

SCHOOL OF PHYSICS
MIDSESSION EXAMINATION

PHYS3114 – Electrodynamics
Term 3, 2019

1. Time allowed – 50 minutes
2. Total number of questions – 3
3. Total marks available – 20
4. Answer ALL questions
5. QUESTIONS ARE NOT OF EQUAL VALUE.
Marks available for each question are shown in the examination paper.
6. Answer all questions in a single book.
7. University-approved calculators may be used.
8. All answers must be written in ink. Except where they are expressly required, pencils may only be used for drawing, sketching or graphical work.
9. This paper may be retained by the candidate.

PHYS3114 — Definitions and Formulae

$$F(s) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x s} dx \quad (1)$$

$$\mathcal{F}\{f(x)\} = 2 \int_0^\infty f_e(x) \cos(2\pi s x) dx - 2i \int_0^\infty f_o(x) \sin(2\pi s x) dx \quad (2)$$

$$V(x', y') = C_1 C_2 \lambda^2 f_o^2 \iint_{\text{bfp}} F\left(\frac{\zeta}{\lambda f_o}, \frac{\eta}{\lambda f_o}\right) e^{-i2\pi(\frac{f_g}{D} x' \frac{\zeta}{\lambda f_o} + \frac{f_o}{D} y' \frac{\eta}{\lambda f_o})} d\left(\frac{\zeta}{\lambda f_o}\right) d\left(\frac{\eta}{\lambda f_o}\right) \quad (3)$$

$$\gamma(P_1, P_2) = \frac{\iint_{\text{source}} I(\zeta, \eta) e^{\frac{-2\pi i}{\lambda}(p\zeta + q\eta)} d\zeta d\eta}{\iint_{\text{source}} I(\zeta, \eta) d\zeta d\eta} \quad (4)$$

$$F(\rho) = 2\pi \int_0^\infty f(r) J_0(2\pi \rho r) r dr \quad (5)$$

$$\frac{m\lambda}{2} = \frac{r_m^2}{2} \left(\frac{1}{r_0} + \frac{1}{b_0} \right) \quad (6)$$

$$\frac{d^2 P}{dt^2} + \gamma \frac{dP}{dt} + \omega_0^2 P = \frac{e^2 N}{m} \left(E + \frac{\nu P}{\epsilon_0} \right). \quad (7)$$

$$\hat{K} - 1 = \frac{\omega_p^2(\omega_{0'}^2 - \omega^2)}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} + i \frac{\omega_p^2 \omega \gamma}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} \quad (8)$$

Theorem	$f(x)$	$F(s)$
Similarity	$f(ax)$	$\frac{1}{ a }F\left(\frac{s}{a}\right)$
Linearity	$\alpha \cdot f(x) + \beta \cdot g(x)$	$\alpha \cdot F(s) + \beta \cdot G(s)$
Shift	$f(x - a)$	$e^{-2\pi ias}F(s)$
Convolution	$f(x) * g(x)$	$F(s)G(s)$
Derivative	$\frac{d}{dx}(f(x))$	$2\pi isF(s)$
Rayleigh	$\int_{-\infty}^{+\infty} f(x) ^2 dx$	$\int_{-\infty}^{+\infty} F(s) ^2 ds$

Table 1: Basic theorems of Fourier theory

Question 1 (4 marks)

Determine the Fourier transform for the function $f(x) = e^{-|x|}$.

Question 2 (10 marks)

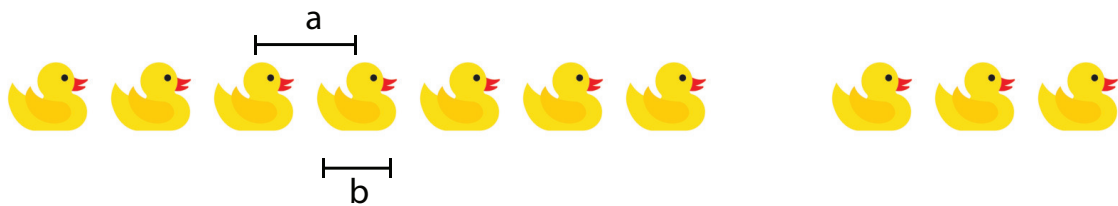
A rectangular aperture, 5 mm in width and 7 mm in height, is placed at a distance, o , in front of a thin lens and illuminated with a monochromatic plane wave ($\lambda = 561$ nm) such that the far-field diffraction pattern is produced at the back focal plane ($f = 40$ cm). An image is formed a further 40 cm past the focus.

The diffraction patterned is filtered with a 2D square array of pinhole apertures (assume delta functions) with a spacing of b . For this problem, assume that the lens aperture is large enough to have a negligible effect on the image.

- (a) Determine the object distance for this lens arrangement.
- (b) Determine an expression for the filtered image.
- (c) Determine the maximum hole spacing that will capture all spatial information contained in the rectangular aperture.
- (d) Discuss the effect of over and under-sampling the diffraction pattern on the resulting image.
- (e) Describe what happens to the image if the pinholes have a finite aperture size.

Question 3 (6 marks)

The following figure contains an image of 11 ducks in a row, but one is missing.



- (a) Write an aperture function that could be used to describe the arrangement.
- (b) Determine the far-field diffraction pattern (in Fourier variables) produced by this aperture function.
- (c) Design a Fourier filter that could be used to return the missing duck to the array. Discuss any limitations on its performance.